INTERNAL FEATURES OF HEART 2

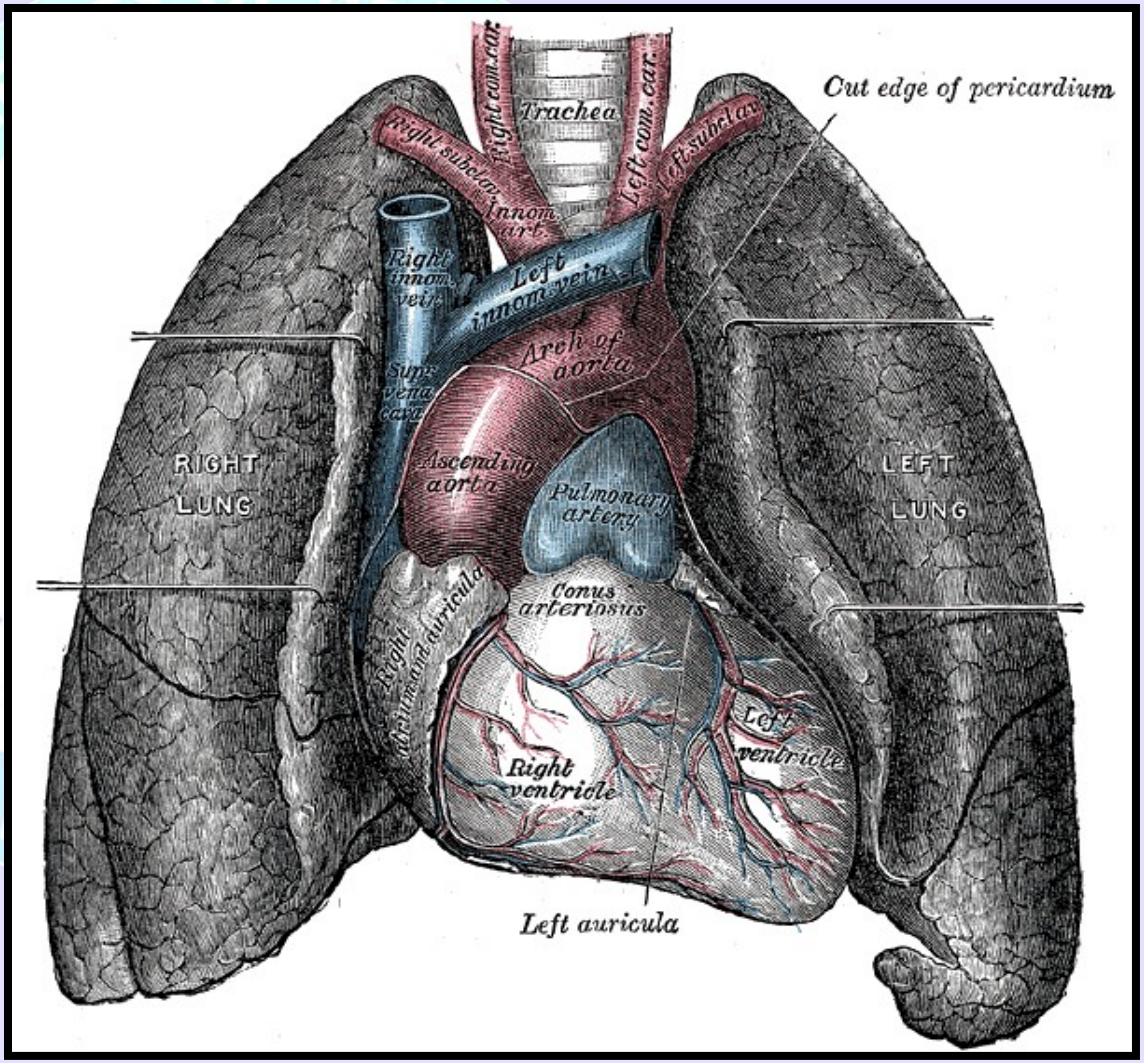
LEARNING OBJECTIVES

At the end of the lecture the student should know the following

- Chambers of the heart
- Internal structures of the heart
- Valves of the heart

HEART

- Heart is a muscular organ.
- Lies between the lungs in the middle mediastinum
- One-third is on the right
- Two third in the left

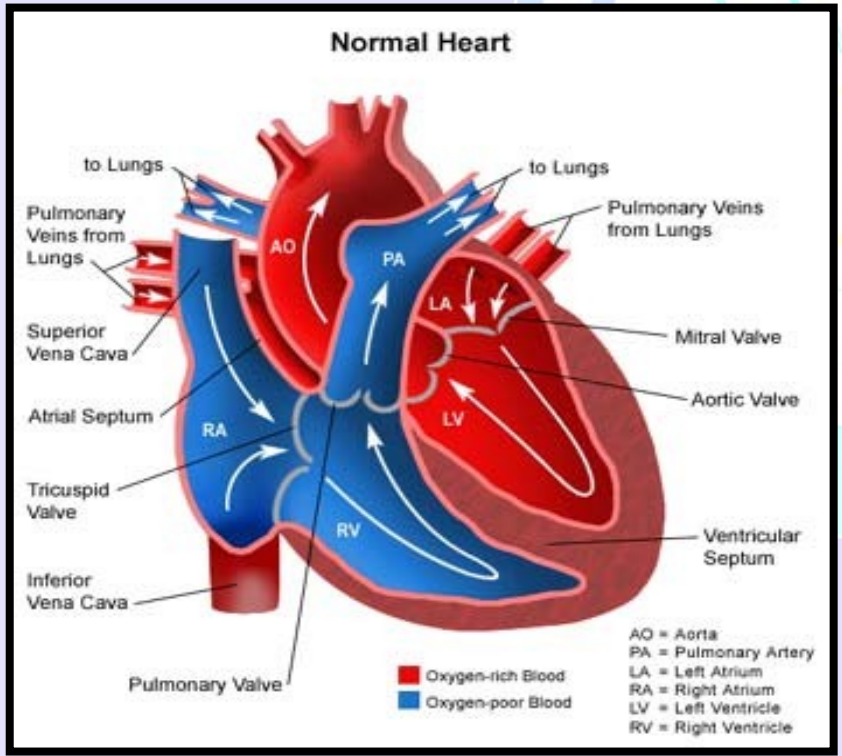


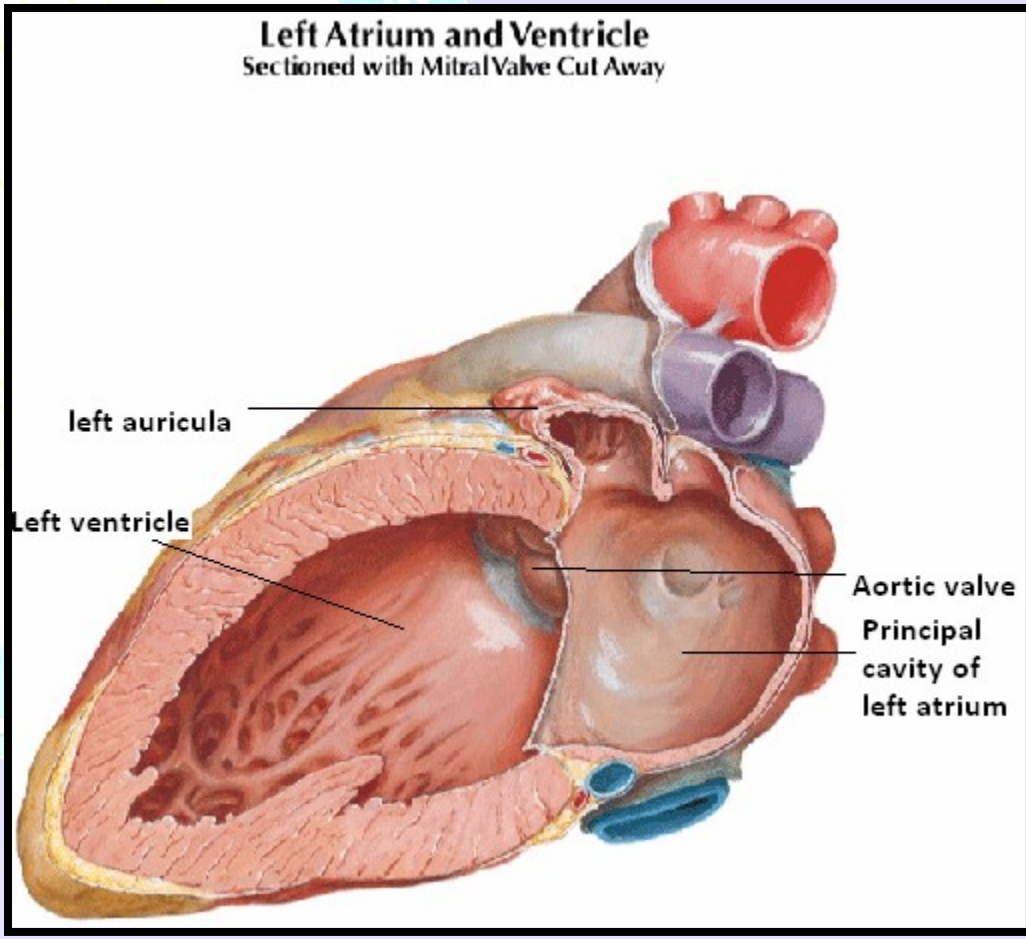
LEFT ATRIUM

- Smaller than the right
- Walls are thick
- Consists of

  Principal cavity

  Auricula





LEFT ATRIUM

- The left atria form most of the base of heart.
- Consist of smooth part and rough part.
- Smooth part : pulmonary veins.
- Rough part : primitive artia..
- Rough part—trabeculated with pectinate muscles.



OPENINGS OF LEFT ATRIUM

- **Pulmonary veins:**

  4 in number

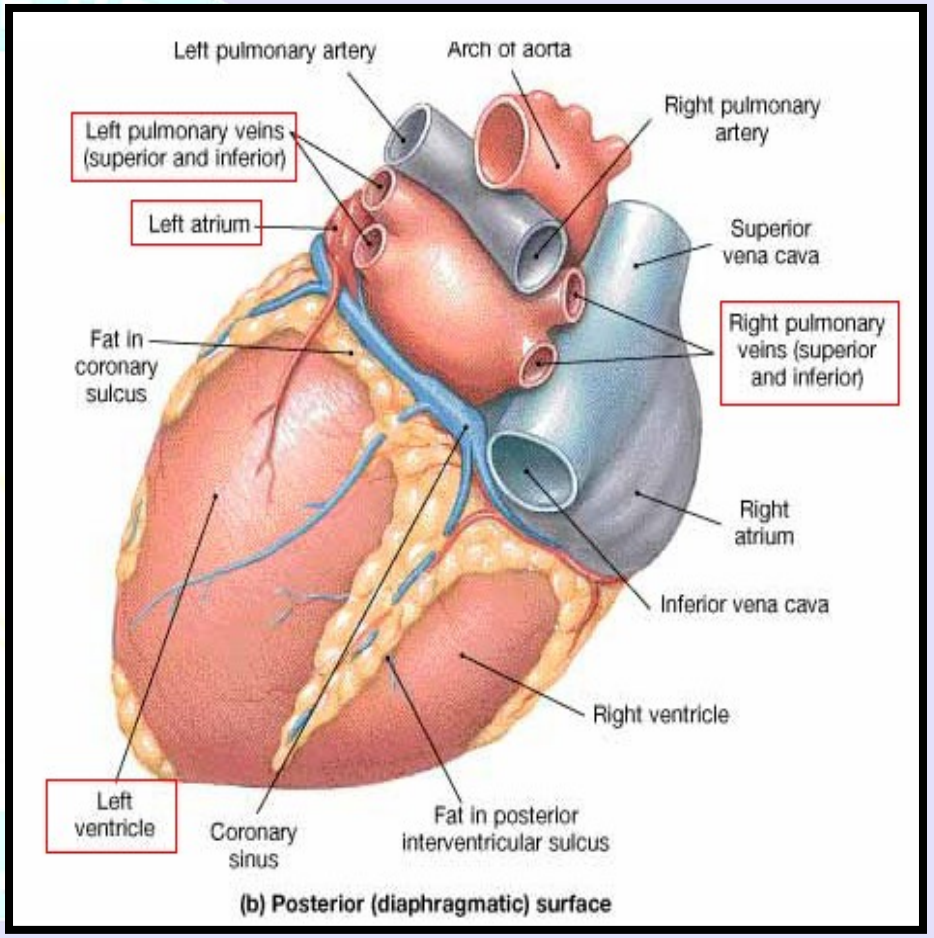
  Opens in the posterior surface

  Has no valves

- **Left Atrioventricular orifice**

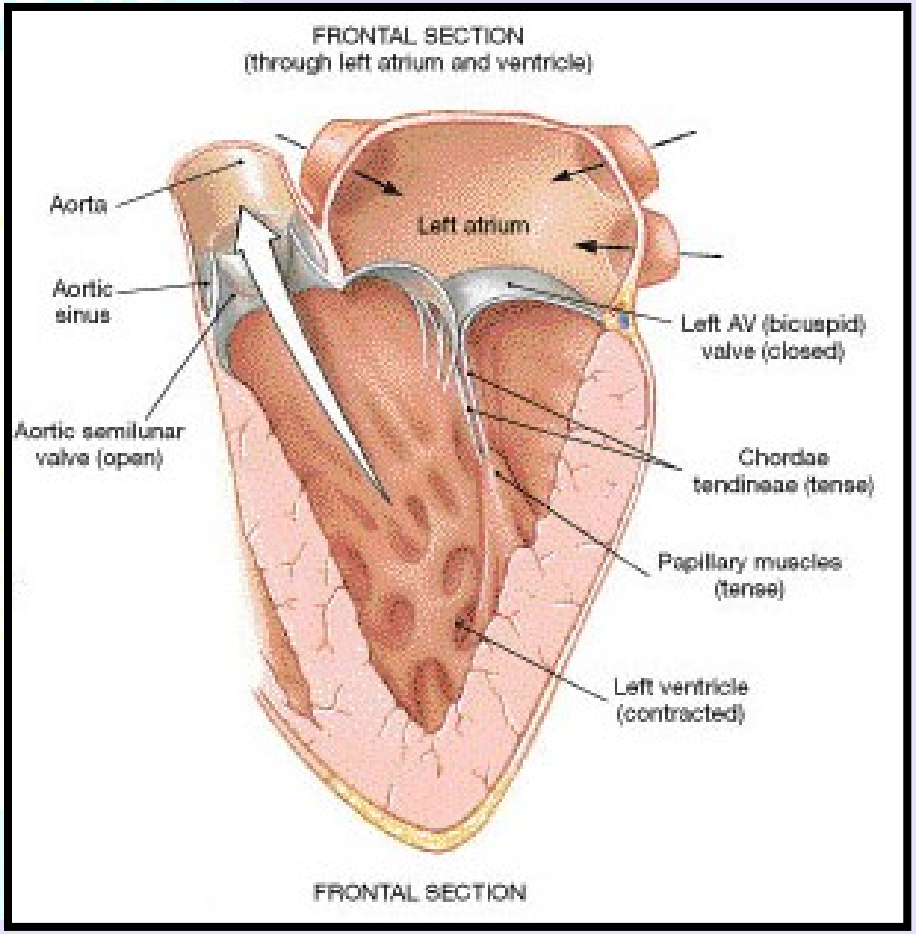
  Between left atrium and ventricle

  Has a mitral valve



LEFT VENTRICLE

- Longer and three times thicker than right
- Conical in shape
- Forms the apex of the heart



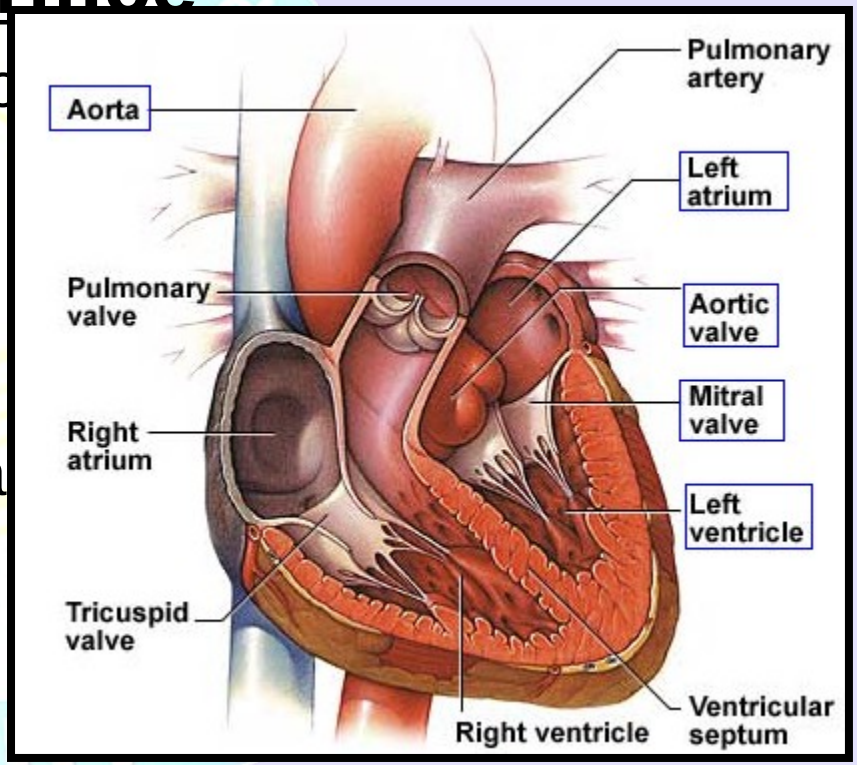
OPENINGS OF LEFT VENTRICLE

Left Atrioventricular orifice

- Below and left to the aortic o
- Has a mitral valve

Aortic orifice

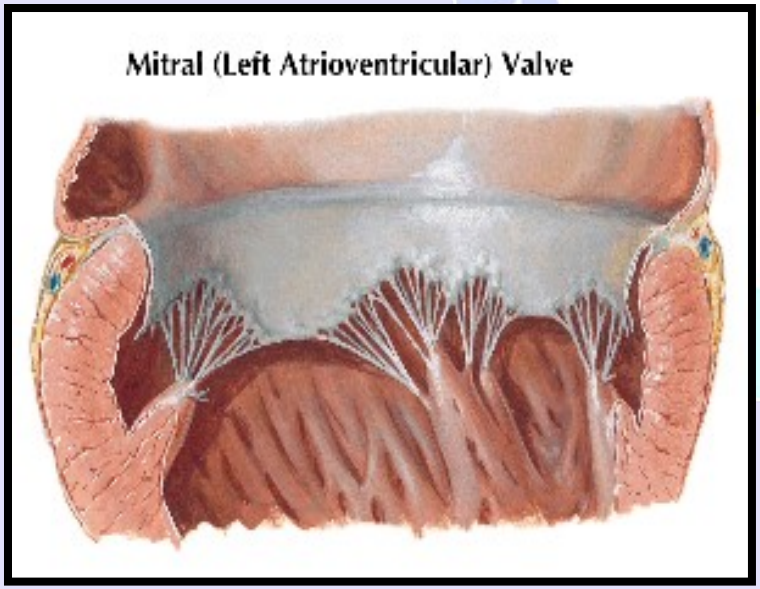
- Circular aperture
- Has an aortic semilunar va

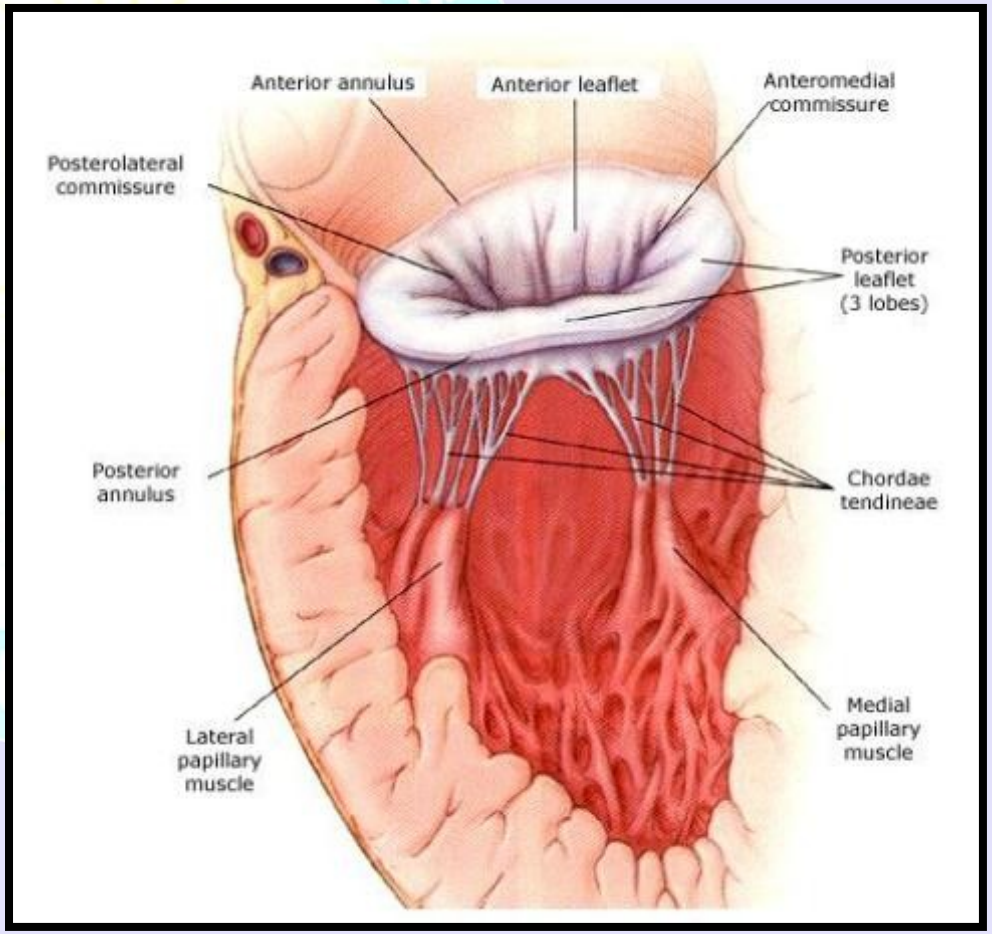


VALVES OF LEFT VENTRICLE

Bicuspid or mitral valve

- Consists of two cusps
- Anterior or aortic cusp
- Posterior cusp

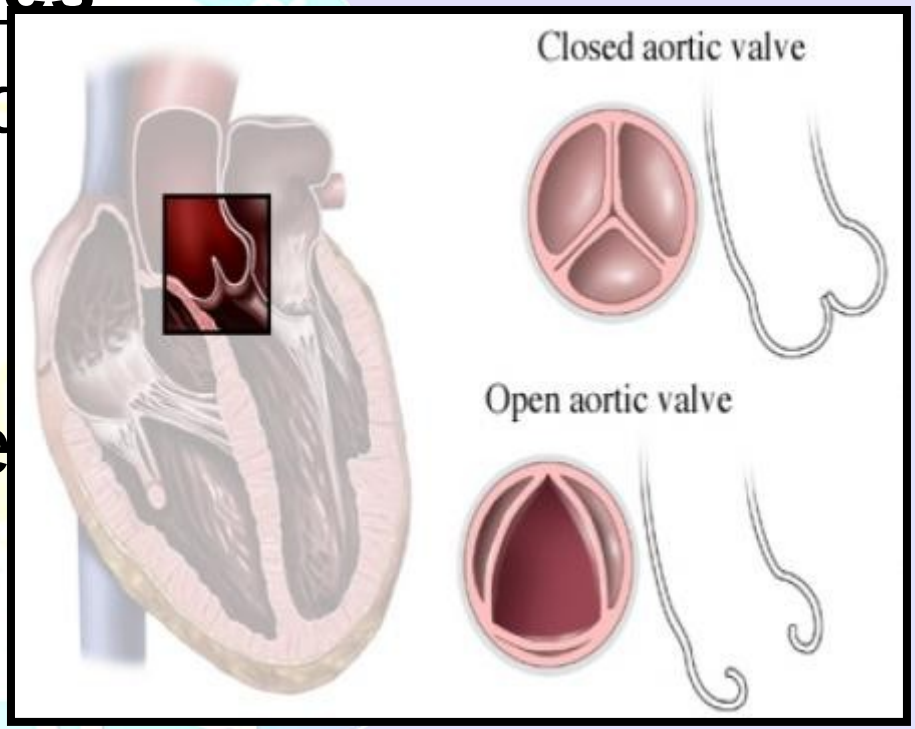




VALVES OF LEFT VENTRICLE

Aortic semilunar valves

- Controls flow of blood
  ventricle to the aorta
- Three in number
- Two anterior and one



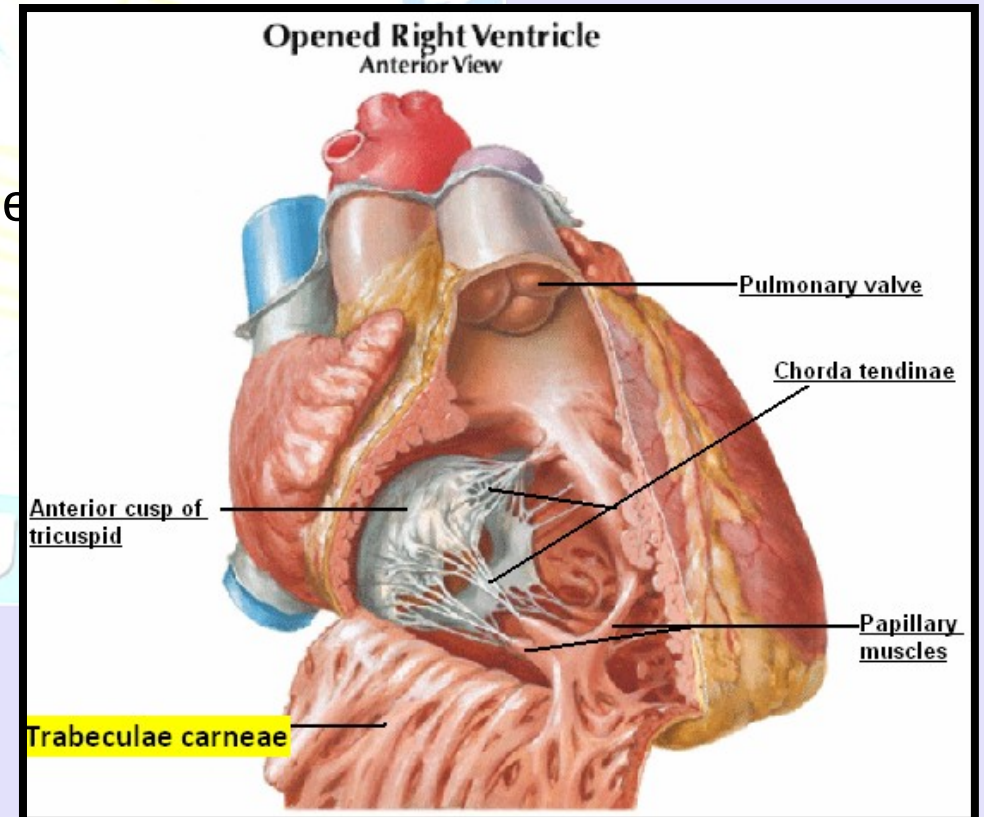
INTERNAL STRUCTURES OF VENTRICLE

- **Trabeculae carneae** (**fleshy beams**)

  They are rounded or irregular muscular columns which project from the whole of the inner surface of the ventricle, with the exception of the conus arteriosus.

- **Three types**

1.Fixed along their entire length
2.Fixed at edges free in the middle
3.Papillary muscles



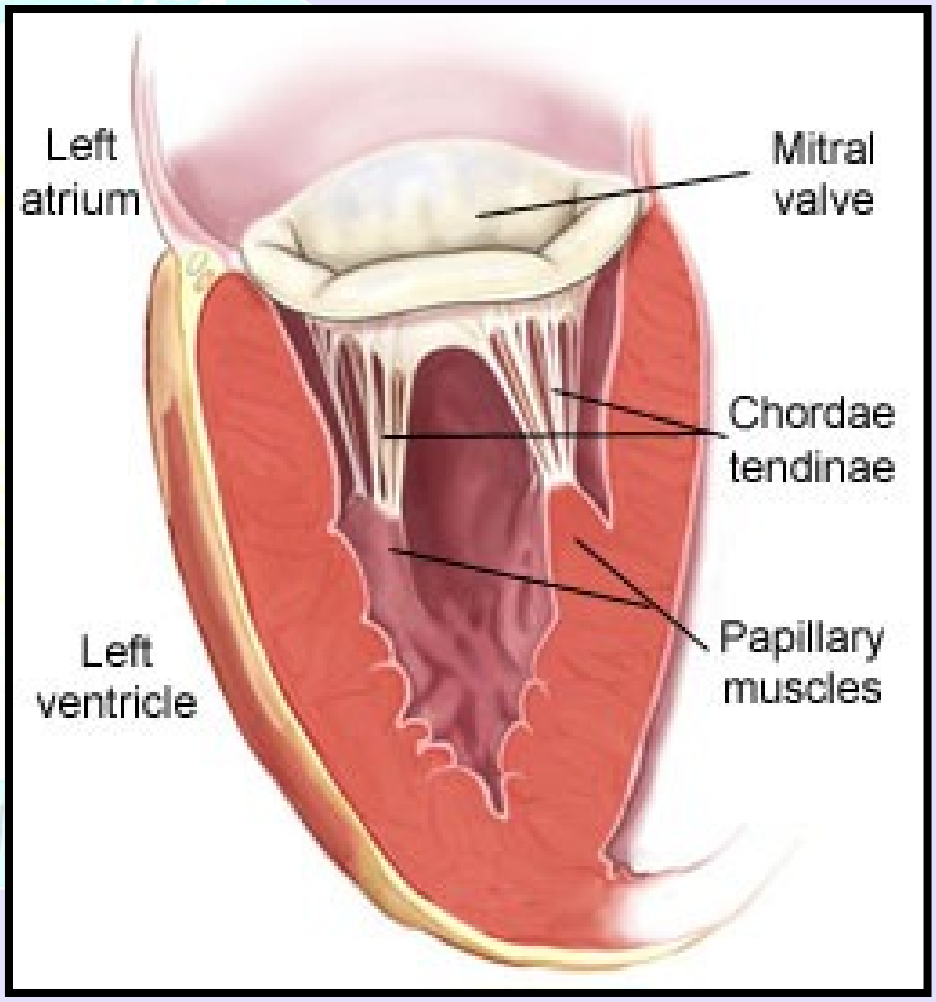
INTERNAL STRUCTURES OF VENTRICLES

- **Papillary muscles**

  Muscles that are attached to the AV valve cusps to limit the movement of the mitral and tricuspid valves

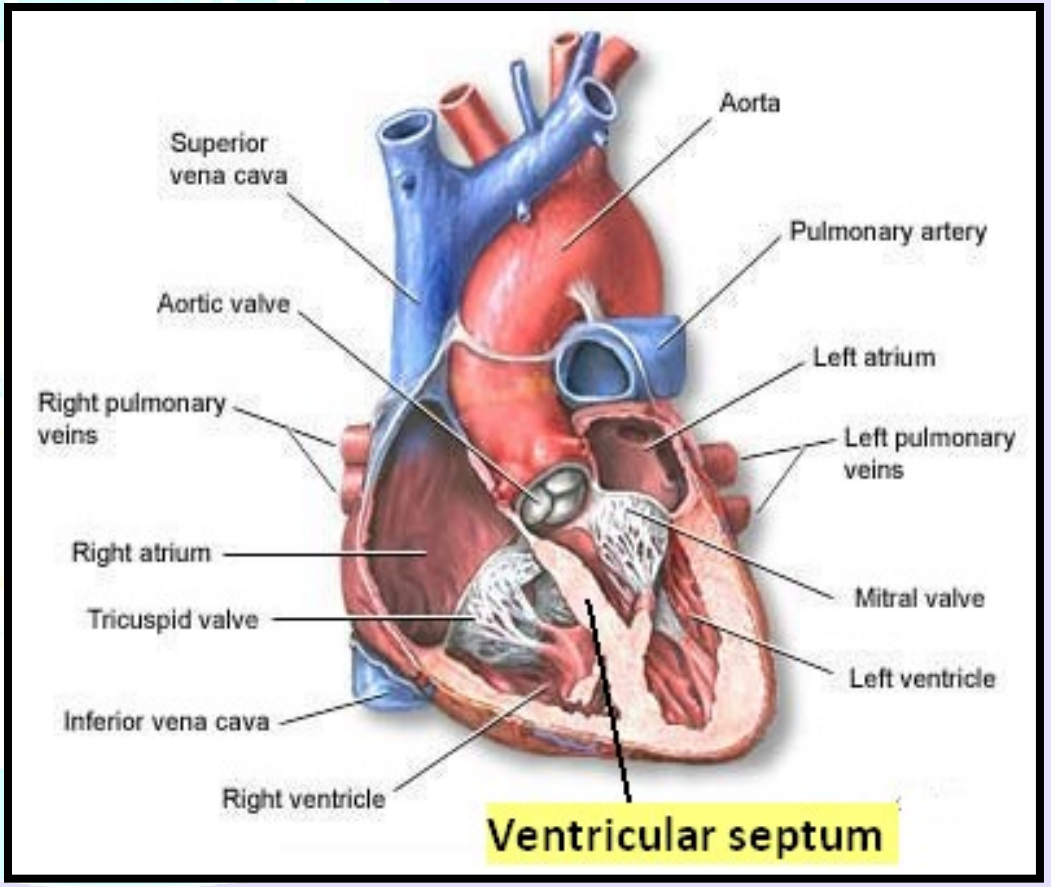
- **Cordae tendineae**

  cord-like tendons that connect the papillary muscles to the tricuspid valve and the mitral valve in the heart.



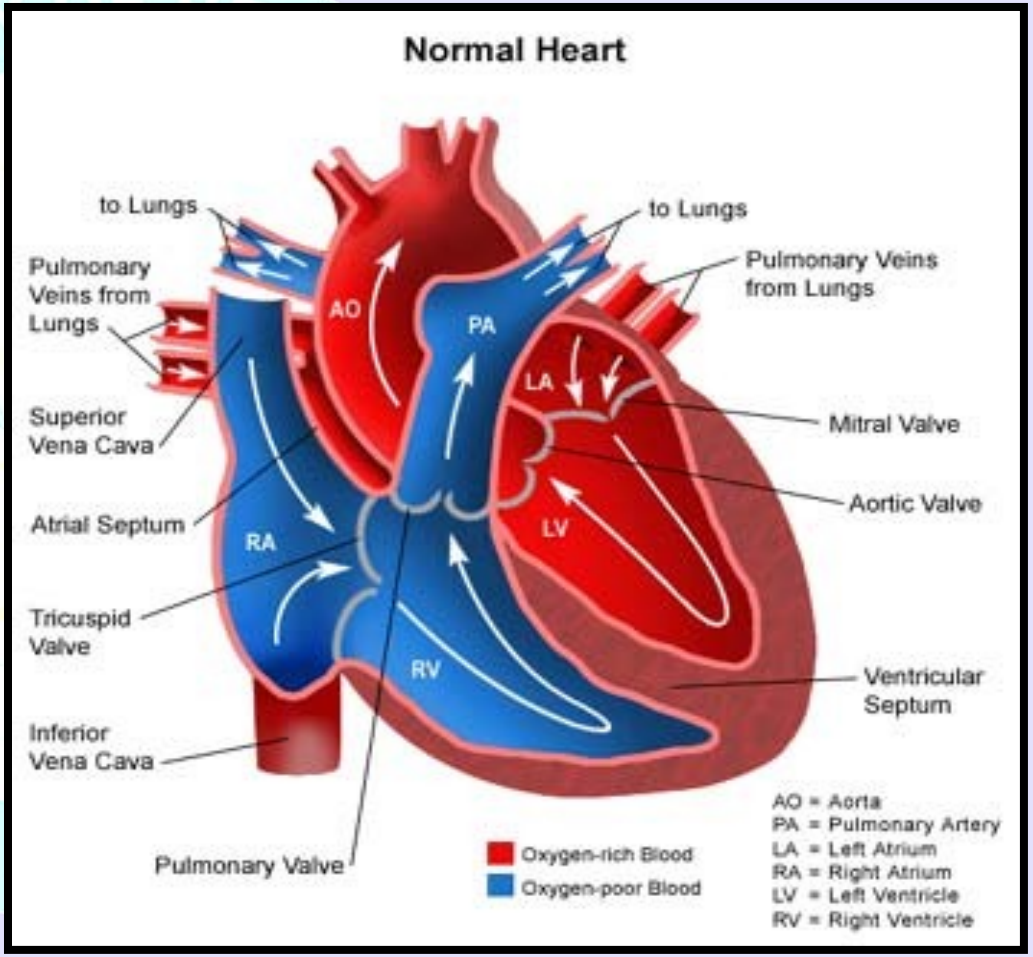
VENTRICULAR SEPTUM

- The ventricular septum is wall separating the lower chamber of the heart from one another
- Directed obliquely backward and to the right, and is curved with the convexity toward the right ventricle:



CHAMBERS OF THE HEART

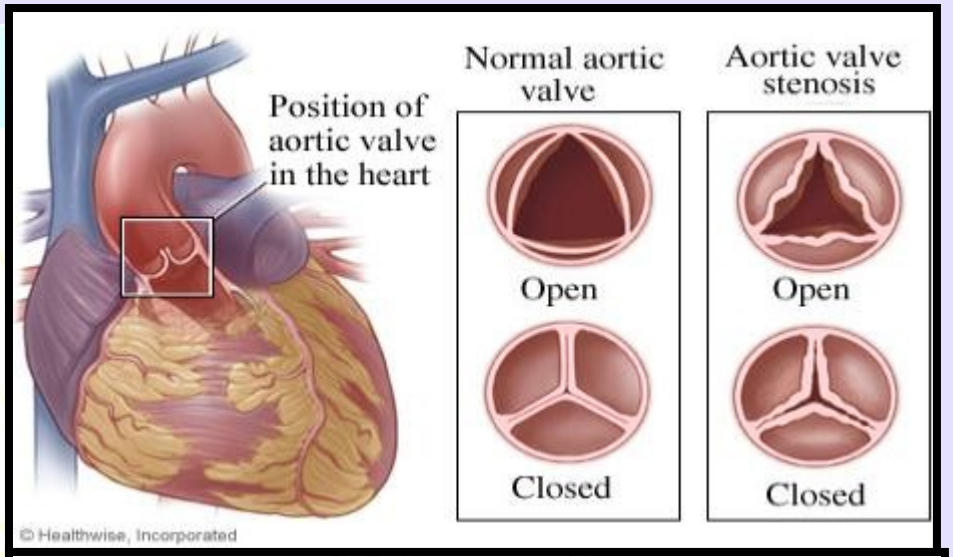
- The right atrium receives blood from the body.
- The right ventricle pumps blood to the lungs.
- The left atrium receives blood from the lungs.
- The left ventricle pumps blood to the rest of the body.



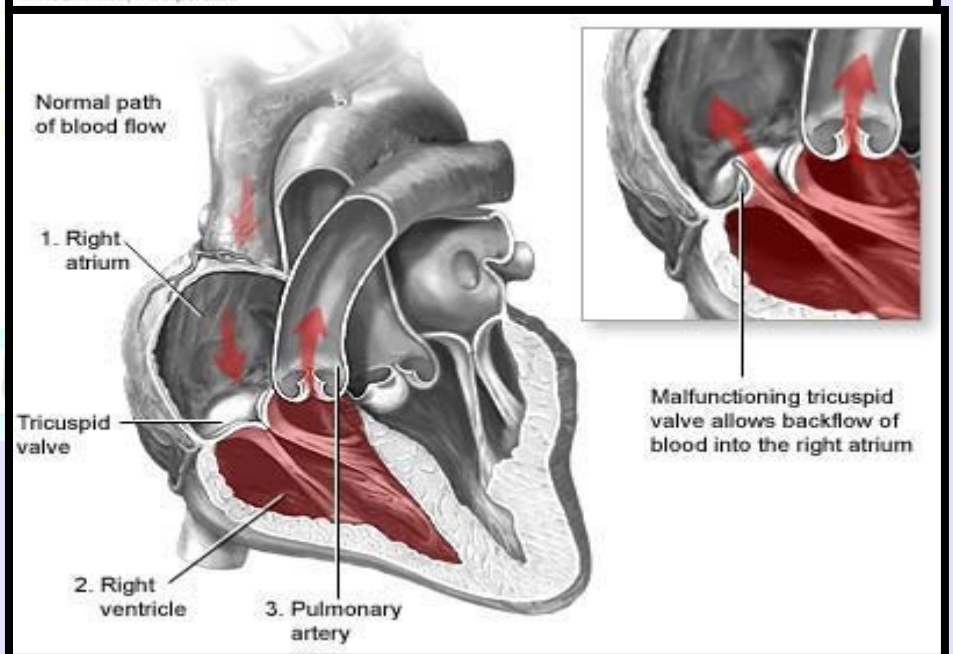
APPLIED ANATOMY

Aortic stenosis

Thickening and hardening of aortic valve which prevents the valve to open properly



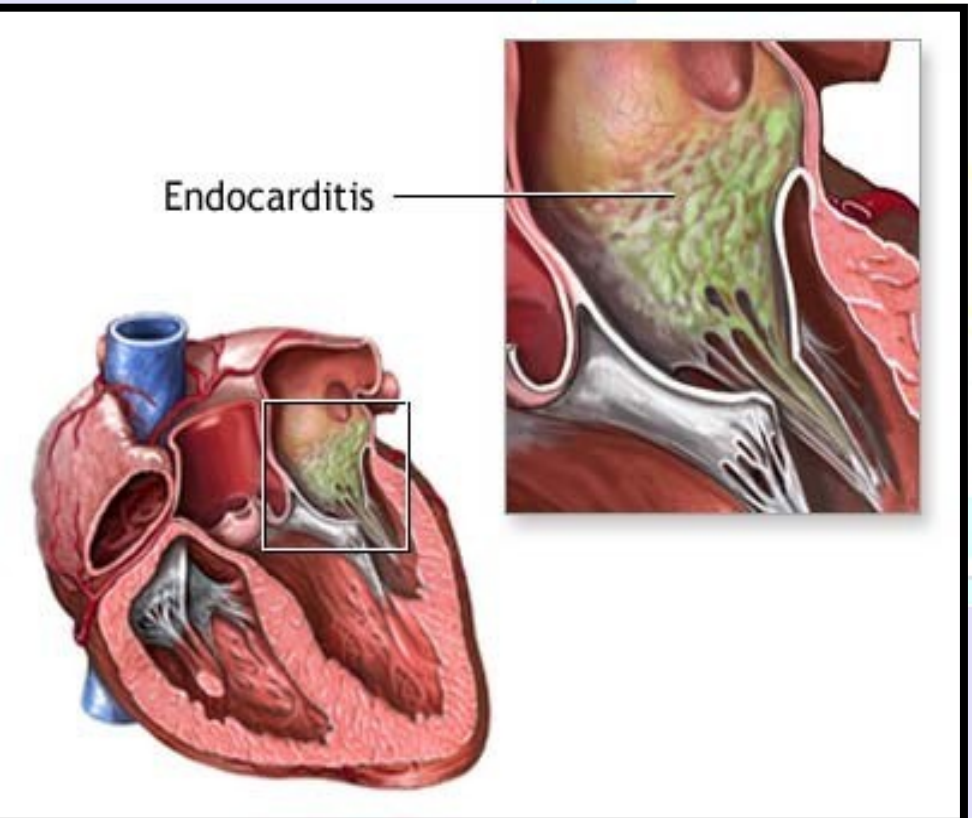
Tricuspid stenosis



APPLIED ANATOMY

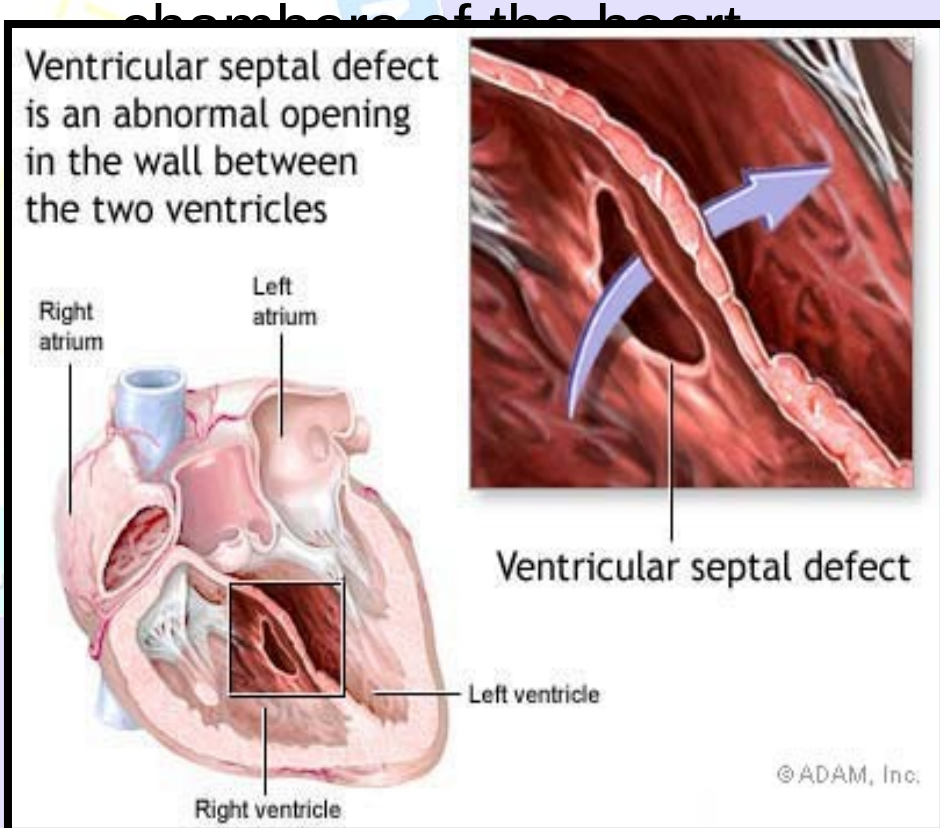
Endocarditis

Infection of the endocardium

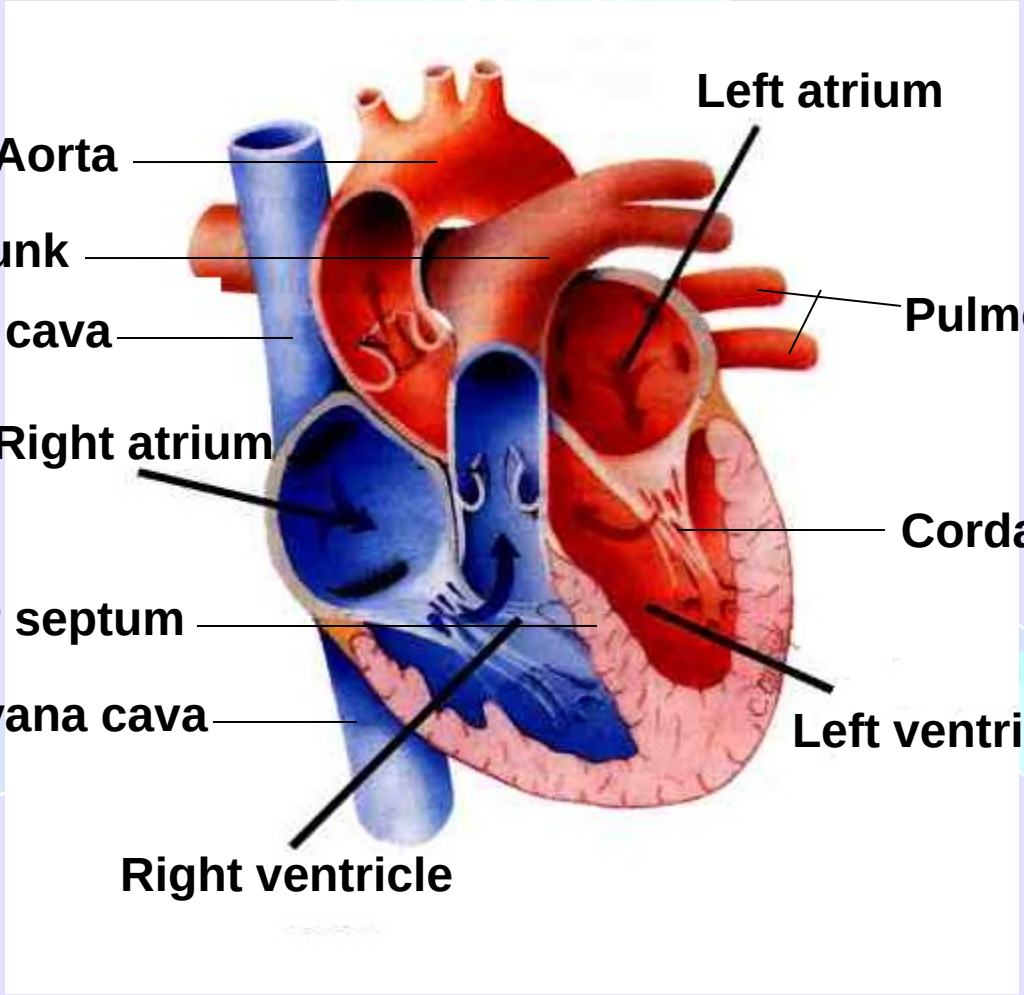


Ventricular septal defect

is a hole (defect) in the wall that separates the lower chambers of the heart



IDENTIFY??


Left atrium
Aorta
Pulmonary trunk
Pulmonary veins
Superior vena cava
Right atrium
Cordae tendinae
Ventricular septum
Inferior vana cava
Left ventricle
Right ventricle

REFERENCES

- **CLINICAL ORIENTED ANATOMY BY K.L.M**

  6TH EDITION



Thank you